

# Efficient Learning of Affine and Rational Dependency LPV Models With Linear Fractional Representation<sup>\*</sup>

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**Abstract:** Identifying control-friendly models of nonlinear systems remains one of the major challenges at the intersection of system identification and control. The *Linear Parameter-Varying* (LPV) framework offers a promising solution, but existing identification methods often rely on model structures with affine scheduling dependency. Instead, this work proposes the use of LPV models with *Linear Fractional Representation* (LFR) admitting a rational scheduling-dependency, capable of modelling complex nonlinear systems with less scheduling variables compared to affine models. This work introduces a direct parameterization to ensure well-posedness of rational LPV-LFR models, which by joint-estimation of an LPV plant, scheduling map and noise process, using only input-output data, is capable of modelling nonlinear systems under complex noise scenarios. Accuracy of the proposed approach is shown on two simulation examples.

**Keywords:** Linear Parameter-Varying System Identification, Linear Fractional Representation, Scheduling Map Estimation

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## 1. INTRODUCTION

As the performance demands of systems in engineering grow, the limitations of the ubiquitous *Linear Time-Invariant* (LTI) assumptions become apparent. Employing the LTI framework to model systems which are often inherently *nonlinear* (NL) results in models with limited accuracy, which subsequently limits achievable performance in (model-based) control.

The *Linear Parameter-Varying* (LPV) framework offers an attractive alternative to the LTI framework. LPV models retain a structure that is linear in the input, but the model parameters can vary as a function of a scheduling variable. Through change in the scheduling variable, LPV models are capable of representing time-varying and nonlinear behaviour. This results in a model class which provides a middle-ground between the limitations of the LTI class and the complexity of nonlinear models. Importantly, the LPV structure enables controller synthesis tools with performance guarantees (see e.g., Mohammadpour and Scherer (2012)). Moreover, the LPV framework enjoys the support of various software toolboxes, implementing

powerful control and modelling approaches (Hjartarson et al., 2015; den Boef et al., 2021).

Traditional LPV identification methods, which require knowledge of the scheduling signal, are often extensions of their LTI counterparts. Notable methods include *Prediction Error Minimization* (PEM) methods (see e.g., Tóth (2010)), and *Subspace Identification* (see e.g., Cox and Tóth (2021)). In recent years methods which jointly identify an LPV model and its scheduling map have been gaining attention. By estimating the scheduling map directly from data, the often significant challenge of modelling the scheduling is bypassed. In particular, *deep-learning*-based identification methods with LPV-*state-space* (SS) models and *Neural Network* (NN) scheduling maps have been shown to efficiently estimate LPV/NL models (Verhoek et al., 2022) and noise processes (Bemporad and Tóth, 2025).

LPV identification methods have been developed for a wide range of model classes, including LPV-*Input-Output* (IO), LPV-SS and LPV-*Linear Fractional Representation* (LFR) classes. However, the LPV-SS and LPV-LFR model classes are most commonly used for controller synthesis. For LPV-SS models, controller synthesis methods require affine dependency on the scheduling variables. Using a simple affine-dependency structure to model complex nonlinear often comes with the trade-off of requiring additional scheduling variables and a complex scheduling map, which in turn can introduce unnecessary conservatism (called

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overbounding) in controller synthesis. Alternatively, we consider the LPV-LFR class of models. This class admits rational dependency on the scheduling variable which can reduce the required dimension and complexity of the scheduling map and thus the effects of overbounding. Importantly, this model class is still suited for controller synthesis (see e.g., Hoffmann and Werner (2014) for a comparison). However, identification of LPV-LFR models corresponding to rational-dependency is a significantly more challenging problem as the rational terms can introduce singularities.

So far, little research on the identification of LPV-LFR models has been performed, and the methods that do exist face significant challenges, such as requiring full-state measurements (Nemani et al. (1995)), being restricted to affine-dependency (Mejari et al. (2019); Bianchi and Sánchez-Peña (2010)), or assuming models cannot be ill-posed (Lee and Poolla (1999), Cheng and Sznajder (2015)). In particular, to the best of the authors' knowledge, no methods to ensure well-posedness in identification of LPV-LFR models with rational scheduling dependency exist. Furthermore, no research on the use of LPV-LFR models in the aforementioned joint scheduling and model identification setting has been performed.

In this work we leverage the efficiency of *deep-learning*-based joint-estimation methods with the expressivity of rational LPV-LFR models to estimate nonlinear models and noise processes. Specifically, as main contribution we provide a direct parameterization of LPV-LFR models that guarantees well-posedness, inspired by direct parameterizations for stability of recurrent models (Revay et al. (2020), Verhoeck et al. (2023)). Subsequently this direct parameterization is incorporated into the nonlinear identification approach proposed in Bemporad and Tóth (2025).

The remainder of this paper first discusses the nonlinear identification setting in Section 2. In Section 3, the LPV-LFR model structure is introduced. Next, in Section 4, well-posedness conditions and a direct-parameterization satisfying the conditions is introduced. This is followed by the introduction of the identification methodology in Section 5. Finally, the proposed method is evaluated on simulation examples in Section 6 after which conclusions are drawn in Section 7.

## 2. IDENTIFICATION SETTING

With the aim of identifying *discrete-time* (DT) nonlinear systems, we consider models with the following innovation-type noise structure:

$$\Sigma_o : \begin{cases} r_{k+1} = f_r(r_k, u_k, e_k), \\ y_k = g_r(r_k, u_k) + e_k, \end{cases} \quad (1)$$

where  $k \in \mathbb{Z}$  is the discrete time, where  $u_k \in \mathbb{U} \subseteq \mathbb{R}^{n_u}$  is the input signal,  $y_k \in \mathbb{Y} \subseteq \mathbb{R}^{n_y}$  is the system output,  $r \in \mathbb{T} \subseteq \mathbb{R}^{n_r}$ ,  $e_k \in \mathbb{E} \subseteq \mathbb{R}^{n_e}$  is the realization of a white noise signal,  $u_k$  is assumed to be quasi-stationary. The state transition function  $f_r : \mathbb{T} \times \mathbb{U} \times \mathbb{E} \rightarrow \mathbb{T}$  and output function  $g_r : \mathbb{T} \times \mathbb{U} \rightarrow \mathbb{Y}$  are considered to be deterministic and nonlinear.

It is assumed that the dynamics (1) can be separated into a deterministic *plant process*,  $G_o$ , according to

$$G_o : \begin{cases} x_{k+1} = f_x(x_k, u_k), \\ y_{o,k} = g_x(x_k, u_k), \end{cases} \quad (2)$$

and a stochastic *noise process*,  $H_o$ , according to

$$H_o : \begin{cases} s_{k+1} = f_s(s_k, x_k, u_k, e_k), \\ v_k = g_s(s_k, x_k, u_k) + e_k, \end{cases} \quad (3)$$

where  $x_k \in \mathbb{X} \subseteq \mathbb{R}^{n_x}$  is the state of the plant,  $s_k \in \mathbb{S} \subseteq \mathbb{R}^{n_s}$  is the state of the noise model and  $v_k \in \mathbb{V} \subseteq \mathbb{R}^{n_v}$  is the output additive form of the noise process.  $f_x : \mathbb{X} \times \mathbb{U} \rightarrow \mathbb{X}$  and  $g_x : \mathbb{X} \times \mathbb{U} \rightarrow \mathbb{Y}$  are the state transition and output functions of the plant and  $f_s : \mathbb{S} \times \mathbb{X} \times \mathbb{U} \times \mathbb{E} \rightarrow \mathbb{S}$  and  $g_s : \mathbb{S} \times \mathbb{X} \times \mathbb{U} \times \mathbb{E} \rightarrow \mathbb{V}$  are the state transition and output functions of the noise representation. A full explanation of the conditions under which this separation is possible is given in Bemporad and Tóth (2025). The output of the deterministic plant,  $y_{o,k}$ , is considered to be observed under output additive noise,  $y_k = y_{o,k} + v_k$ , which is defined by the stochastic noise process according to (3).

Through structural constraints on (3), the combined plant and noise structure is capable of representing a wide range of NL noise scenarios, including *Nonlinear AutoRegressive model with eXogenous input* (NARX), *Nonlinear Output-Error* (NOE), *Nonlinear AutoRegressive Moving Average model with eXogenous input* (NARMAX) and *Nonlinear Box-Jenkins* (NBJ). This provides a systematic way of modelling nonlinear systems which is similar to methods popular for LTI systems.

By substituting  $e_k = v_k - g_s(s_k, x_k, u_k)$  into the equations of (3), a direct characterization of the inverse noise process,  $H_o^{-1}$  can be made:

$$H_o^{-1} : \begin{cases} s_{k+1} = \bar{f}_s(s_k, x_k, u_k, v_k), \\ e_k = \bar{g}_s(s_k, x_k, u_k) + v_k, \end{cases} \quad (4)$$

where  $\bar{f}_s(s_k, x_k, u_k, v_k) = f_s(s_k, x_k, u_k, v_k - g_s(s_k, x_k, u_k))$  and  $\bar{g}_s(s_k, x_k, u_k) = -g_s(s_k, x_k, u_k)$ . Allowing for direct computation of the nonlinear one-step ahead predictor, resulting in the following state-space representation:

$$x_{k+1} = f_x(x_k, u_k), \quad (5a)$$

$$s_{k+1} = \bar{f}_s(s_k, x_k, u_k, y_k - g_x(x_k, u_k)), \quad (5b)$$

$$\hat{y}_k = g_x(x_k, u_k) - \bar{g}_s(s_k, x_k, u_k). \quad (5c)$$

## 3. LPV-LFR MODEL STRUCTURES FOR IDENTIFICATION

### 3.1 Linear Fractional Representation

The *Linear Fractional Representation* is a general model representation that allows one to model dynamical systems with a wide range of properties, such as nonlinearities and uncertainty, in a unified manner. It originally gained popularity in the field of robust control (Zhou et al. (1996)) where it is used to model uncertain systems. For LPV systems it is instead used to incorporate the parameter-varying nature. With an LFR, LPV systems are represented by an interconnection between a nominal system  $M$  and its parameter-varying components represented by a matrix function  $\Delta(p)$  with linear dependence on the scheduling signal  $p$ .

Let us define an LPV-LFR as a pair  $\{M, \Delta(p)\}$  where the interconnection in discrete time follows the standard form:

$$M : \begin{cases} \begin{bmatrix} x(k+1) \\ z(k) \\ y(k) \end{bmatrix} = \begin{bmatrix} A_x & B_w & B_u \\ C_z & D_{zw} & D_{zu} \\ C_y & D_{yw} & D_{yu} \end{bmatrix} \begin{bmatrix} x(k) \\ w(k) \\ u(k) \end{bmatrix}, \\ w(k) = \Delta(p(k))z(k), \end{cases} \quad (6)$$

where  $w(k), z(k) \in \mathbb{R}^{n_w}$  are the latent variables,  $A_x, \dots, D_{yu}$  are real matrices of appropriate dimensions, and  $p(k) \in \mathbb{P} \subseteq \mathbb{R}^{n_p}$  is the scheduling variable. The matrix function  $\Delta(p(k)) : \mathbb{P} \rightarrow \mathbb{R}^{n_w \times n_w}$ , commonly referred to as the Delta-block, is defined as:

$$\Delta(p(k)) = \begin{bmatrix} p_1(k)I_{\eta_1} & & 0 \\ & \ddots & \\ 0 & & p_{n_p}(k)I_{\eta_{n_p}} \end{bmatrix}, \quad (7)$$

where  $\eta$  is a vector of integers denoting the number of repetitions per scheduling variable and where  $I_{\eta_1}, \dots, I_{\eta_{n_p}}$  are identity matrices of appropriate size to realize the correct total dimension of  $\Delta(p(k))$ . This interconnection is equivalent to a LPV-SS representation with rational dependency on  $p(k)$ :

$$\begin{bmatrix} \dot{x}(k) \\ y(k) \end{bmatrix} = \begin{bmatrix} \mathcal{A}(p(k)) & \mathcal{B}(p(k)) \\ \mathcal{C}(p(k)) & \mathcal{D}(p(k)) \end{bmatrix} \begin{bmatrix} x(k) \\ u(k) \end{bmatrix}, \quad (8)$$

where  $\mathcal{A}(p(k)), \mathcal{B}(p(k)), \mathcal{C}(p(k)), \mathcal{D}(p(k))$  are now matrix functions of  $p(k)$ .

This is achieved by eliminating the latent variables  $w(k)$  and  $z(k)$  in (6) by substitution of

$$z(k) = (I - D_{zw}\Delta(p(k)))^{-1}(C_z x(k) + D_{zu}u(k)). \quad (9)$$

Which requires  $(I - D_{zw}\Delta(p(k)))$  to be non-singular. Thus, we define the following:

*Definition 1.* The LPV-LFR model defined by the pair  $\{M, \Delta(p)\}$  according to (6) is *well-posed* if  $I - D_{zw}\Delta(p(k))$  is non-singular, i.e.,  $\det(I - D_{zw}\Delta(p(k))) \neq 0$  for all possible realizations of the scheduling variable, i.e.,  $p(k) \in \mathbb{P}$ .

As this representation admits rational LPV-SS models, it is a generalization of commonly used affine LPV-SS models. Notably, affine-dependency models are represented by taking (6) with  $D_{zw}=0$ , resulting in a model set which is always well-posed. Furthermore, given an LPV-SS model with affine dependency an equivalent LPV-LFR realization can easily be recovered using *Singular Value Decomposition* (SVD) methods (Zhou et al., 1996).

LPV-LFR models can be used to represent LPV systems by considering  $p(k)$  to be an exogeneous signal. Nonlinear systems such as  $G_o$  in (2) require defining  $p(k)$  according to a *scheduling map*  $\psi : \mathbb{R}^{n_x} \times \mathbb{R}^{n_u} \rightarrow \mathbb{R}^{n_p}$  corresponding to the LPV embedding of the original nonlinear system.

### 3.2 Identification structure

We aim to identify the data-generating plant  $G_o$  and noise process  $H_o$  based on a dataset  $\mathcal{D}_N$  in the form of parameterized surrogate models. Specifically, we consider *self-scheduled* LPV-LFR models, requiring the joint-estimation of LPV model and scheduling map, similar to Verhoeck et al. (2022).

We parameterize the surrogate model for the plant (2) in an LPV-LFR form according to:

$$G_\theta : \begin{cases} \begin{bmatrix} \hat{x}_{k+1} \\ \hat{z}_k \\ \hat{y}_k \end{bmatrix} = \begin{bmatrix} A_x & B_w & B_u \\ C_z & D_{zw} & D_{zu} \\ C_y & D_{yw} & D_{yu} \end{bmatrix} \begin{bmatrix} \hat{x}_k \\ \hat{w}_k \\ u_k \end{bmatrix}, \\ \hat{w}_k = \Delta_x(\hat{p}_k)\hat{z}_k. \end{cases} \quad (10)$$

We directly characterize the surrogate model of the inverse noise process (3) as:

$$H_\theta^{-1} : \begin{cases} \begin{bmatrix} \hat{s}_{k+1} \\ \hat{\zeta}_k \\ \hat{e}_k \end{bmatrix} = \overbrace{\begin{bmatrix} A_s & B_\omega & B_v \\ C_\zeta & D_{\zeta\omega} & 0 \\ C_e & D_{e\omega} & I \end{bmatrix}}^{S_\theta} \begin{bmatrix} \hat{s}_k \\ \hat{\omega}_k \\ \hat{v}_k \end{bmatrix}, \\ \hat{\omega}_k = \Delta_s(\hat{p}_k)\hat{\zeta}_k. \end{cases} \quad (11)$$

where  $\zeta, \omega \in \mathbb{R}^{n_\omega}$  denote the latent variables of the noise process and  $S_\theta$  represents the LTI part of the LFR. Ensuring the noise process retains the innovation-type representation requires setting  $D_{ev} = I$ , and  $D_{\zeta v} = 0$  or  $D_{e\omega} = 0$ . In the remainder of this work, we construct  $H_\theta^{-1}$  according to (11) with  $D_{\zeta v} = 0$ . One benefit of using an LPV-LFR model is that the inverse of an LFR is also an LFR. If the original noise process  $H_\theta$  has to be recovered this is easily achieved by inverting the inputs and outputs of  $S_\theta$ , resulting in

$$\bar{S}_\theta : \begin{bmatrix} A_s - B_v D_{ev}^{-1} C_e & B_\omega - B_v D_{ev}^{-1} D_{e\omega} & B_v D_{ev}^{-1} \\ C_\zeta - D_{\zeta v} D_{ev}^{-1} C_e & D_{\zeta\omega} - D_{\zeta v} D_{ev}^{-1} D_{e\omega} & D_{\zeta v} D_{ev}^{-1} \\ -D_{ev}^{-1} C_v & -D_{ev}^{-1} D_{e\omega} & D_{ev}^{-1} \end{bmatrix} \quad (12)$$

under the condition that  $D_{zv}$  is non-singular.

Next, we consider the scheduling signal  $\hat{p}_k$  to be defined by a scheduling map  $\hat{\psi}_\theta(x_k, u_k, d_k)$ , where  $d_k \in \mathbb{R}^{n_d}$  is a known exogenous signal. The inclusion of  $d_k$  allows for full or partial external scheduling. To be able to capture the set of data-generating systems described in (1), the scheduling map is required to be a general function approximator. To achieve this, we propose the use of a *Feedforward Neural Network* (FNN) parameterization of the scheduling map. Typically the FNN is extended with a linear bypass layer, resulting in a one-block ResNet (He et al., 2016). This bypass aids in the learning of linear dependencies on  $x_k, u_k$  and  $d_k$  if present. The authors suggest the use of a ResNet with two hidden layers, limited amount of neurons per layer (6-12) and hyperbolic tangent activation functions.

## 4. ENSURING WELL-POSEDNESS

### 4.1 Well-posedness conditions

As we have shown in (9), for an LPV-LFR to be well-posed, we require  $I - D_{zw}\Delta(p(k))$  to be non-singular for all possible realizations of the scheduling variables, i.e.,  $p(k) \in \mathbb{P}$ . We propose a novel method for guaranteeing well-posedness of LPV-LFR models, we begin by establishing key assumptions and conditions that play a role in this result.

*Assumption 2.* The matrix function  $\Delta(p(k))$  has a diagonal structure.

*Assumption 3.* The scheduling set  $\mathbb{P}$  is contained in the closed  $n_p$ -dimensional  $\ell_\infty$  unit ball, i.e.,

$$\mathbb{P} \subseteq \{p : \mathbb{Z} \rightarrow \mathbb{R}^{n_p} \mid \|p(k)\|_\infty \leq 1 \forall k \in \mathbb{Z}\}. \quad (13)$$

It is important to note that Assumption 3 is not restrictive. Given an arbitrary, well-posed, LFR  $\{G, \Delta(p)\}$ , under

Assumption 2 it is trivial to retrieve an equivalent LFR defined by the pair  $\{\bar{M}, \bar{\Delta}(p)\}$  which satisfies Assumption 3 by scaling and shifting the scheduling variables (see e.g., Scherer and Weiland (2015)). For models with NN scheduling map this can be enforced through the incorporation of an activation function on the output layer, such as the tanh function.

*Definition 4.* The *spectral radius*  $\rho$  of a matrix  $A \in \mathbb{R}^{n \times n}$  with eigenvalues  $\lambda_1, \dots, \lambda_n$  is defined as:

$$\rho(A) = \max\{|\lambda_1|, \dots, |\lambda_n|\}. \quad (14)$$

*Condition 5.* The spectral radius  $\rho$  of  $D_{zw}$  is strictly smaller than 1, i.e.,

$$\rho(D_{zw}) < 1. \quad (15)$$

These lead to the following theorem.

*Theorem 6.* If Assumptions 2, 3, and Condition 5 hold, the LPV-LFR model (6) is well-posed.

**Proof.** From Assumptions 2 and 3 it follows that  $\rho(\Delta(p(k))) \leq 1$ ,  $\forall p(k) \in \mathbb{P}$ . Additionally, under Assumption 2, it holds that  $\rho(D_{zw}\Delta(p(k))) \leq \rho(D_{zw})\rho(\Delta(p(k)))$ . Thus, it follows that  $I - D_{zw}\Delta(p(k)) \succ 0$  for all combinations of  $p(k) \in \mathbb{P}$ . This implies  $\det(I - D_{zw}\Delta(p(k))) > 0$ ,  $\forall p(k) \in \mathbb{P}$ , satisfying the well-posedness condition.

#### 4.2 Well-posedness by direct parameterization

To ensure Condition 5 is satisfied, we define

$$D_{zw} = e^{-N}, \quad N \succ 0. \quad (16)$$

As the matrix exponential maps negative definite matrices to matrices  $D_{zw} \prec I$ . Based on (16) we propose direct parameterization satisfying  $N \succ 0$ , enabling unconstrained optimization.

To guarantee Condition 5 through  $D_{zw}$  defined according to (16) with  $N \succ 0$ , we use the parameterization

$$N = \Psi(D_A^\top D_A + D_B - D_B^\top + \epsilon I), \quad (17)$$

where the matrix  $\Psi = \text{diag}(e^{D_d})$  such that  $N$  is constructed with the free variables  $D_A \in \mathbb{R}^{n_w \times n_w}$ ,  $D_B \in \mathbb{R}^{n_w \times n_w}$  and  $D_d \in \mathbb{R}^{n_w}$  and a constant  $0 < \epsilon \ll 1$ . By combining a symmetric positive-semi-definite term  $D_A^\top D_A$ , a skew-symmetric term  $D_B - D_B^\top$  and a strictly positive definite term  $\epsilon I$ , with positive scaling by  $\Psi$ ,  $N$  is guaranteed to be strictly positive-definite by construction. The positive scaling term  $\Psi$  is shown in Revay et al. (2020) to extend the feasible domain of  $N$ . Additionally, it aids in counteracting potential poor scaling of variables which can be introduced through the transformation (16).

When considering the use of LPV-LFR models in identification, the parameterization of  $N$  according to (17) enables unconstrained optimization without risk of obtaining ill-posed models. By substituting the direct free parameterization of  $D_{zw}$  with the transformed free variables in  $D_A, D_B$  and  $D_d$  for constructing  $D_{zw}$ , the number of parameters is increased from  $n_w^2$  to  $2n_w^2 + n_w$ . However, as noted in Winston and Kolter (2020), by populating only the lower triangular elements of  $D_A$  and the strictly upper triangular elements of  $D_B$ , this overparameterization is greatly reduced, now requiring only  $n_w^2 + n_w$  elements,  $n_w$  variables more than populating  $D_{zw}$  directly.

## 5. IDENTIFICATION METHOD

### 5.1 Parameter Estimation

We aim to estimate a parametric surrogate model using LFR parameterizations according to (10), (11) and given a dataset  $\mathcal{D}_N$  by minimizing the criterion:

$$V_N^{\text{fit}}(\theta, \mathcal{D}_N) = \frac{1}{N} \sum_{k=0}^{N-1} \|\hat{e}_{k|\theta}\|_2^2, \quad (18)$$

where  $\hat{e}_{k|\theta} = y_k - \hat{y}_{k|\theta}$  represents the prediction error under the parameters  $\theta$ .

The parameter vector  $\theta$  is comprised of the combined parameter vectors of the plant, noise model and scheduling map, i.e.,  $\theta = [\theta_G^\top \theta_H^\top \theta_\psi^\top]^\top$ . The minimization of (18) matches the classical PEM criterion which aims to minimize the *Mean Squared Error* (MSE) of the predictions.

By combining the PEM criterion (18), one-step-ahead prediction equations (5), and the LPV-LFR model structure, the identification problem is formulated as the following optimization problem:

$$\begin{aligned} \underset{\theta}{\text{argmin}} \quad & V_N^{\text{fit}}(\theta, \mathcal{D}_N) \\ \text{s.t.} \quad & x_0 = \theta_{x_0} \\ & s_0 = \theta_{s_0} \\ & \hat{p}_k = \psi(\hat{x}_k, u_k, d_k, \theta) \\ & \hat{x}_{k+1} = \mathcal{A}_x(\Delta_x(\hat{p}_k), \theta) \hat{x}_k + \mathcal{B}_x(\Delta_x(\hat{p}_k), \theta) u_k \\ & \hat{y}_k = \mathcal{C}_x(\Delta_x(\hat{p}_k), \theta) \hat{x}_k + \mathcal{D}_x(\Delta_x(\hat{p}_k), \theta) u_k \\ & \hat{s}_{k+1} = \mathcal{A}_s(\Delta_s(\hat{p}_k), \theta) \hat{s}_k + \mathcal{B}_s(\Delta_s(\hat{p}_k), \theta) (y_k - \hat{y}_k) \\ & \hat{e}_k = \mathcal{C}_s(\Delta_s(\hat{p}_k), \theta) \hat{s}_k + (y_k - \hat{y}_k) \end{aligned} \quad (19)$$

To solve this optimization problem, we leverage the optimization approach described in Bemporad (2025) based on JAX (Bradbury et al. (2018)). In this gradient-based method, local optimization is performed in two steps. First, a fixed number of iterations is taken using the Adam algorithm, followed by a number of iterations using the *bound-constrained limited-memory Broyden-Fletcher-Goldfarb-Shanno* (L-BFGS-B) algorithm (Liu and Nocedal (1989)). The use of Adam, through the incorporation of *momentum*, aids in escaping local minima at the early iterations of the optimization process, providing a better point of initialization for the L-BFGS-B optimizer, which subsequently converges accurately to the final minimum of the optimization process. This two-step approach has been shown to be computationally efficient for the identification of NL-SS systems (Bemporad (2025)) as well as self-scheduled LPV-SS models (Bemporad and Tóth (2025)), particularly when compared with methods using Adam exclusively.

### 5.2 Parameter Initialization

When starting the optimization procedure, initial values for all elements in the parameter vector are required. This is done a pseudo-random manner. The elements of the plant are initialized as follows:

$$\begin{aligned} A_x &= 0.5I, & B_w &= \mathbf{0}, \\ B_u &\sim \mathcal{U}(-0.1, 0.1), & C_z &\sim \mathcal{U}(-0.1, 0.1), \\ D_{zu} &\sim \mathcal{U}(-0.1, 0.1), & C_y &\sim \mathcal{U}(-0.1, 0.1), \\ D_{yw} &= \mathbf{0}, & D_{yu} &= \mathbf{0}. \end{aligned} \quad (20)$$

If affine LPV-LFR models are used,  $D_{zw}$  is left out of the parameter vector, fixed at a zero matrix. For well-posed rational LPV-LFR models, we construct the  $D_{zw}$  matrix by drawing lower triangular elements from a uniform distribution  $\mathcal{U}(0, 0.1)$ , the strictly upper triangular elements of  $D_B^x$  follow the normal distribution  $\mathcal{N}(1, 1)$ , finally  $D_d^x = \mathbf{0}$  is chosen to initialize the remaining parameter vector.

The LFR parameters of the noise model are initialized similarly to their plant equivalents with a change in the scaling of  $A_s$  and the omission of  $D_{\zeta v}$  and  $D_{ev}$  as these are fixed at a zero matrix and identity matrix respectively, to ensure the noise process is monic. The remaining parameters are initialized using the following strategy:

$$\begin{aligned} A_s &= 0.1I, & B_\omega &= \mathbf{0}, \\ B_v &\sim \mathcal{U}(-0.1, 0.1), & C_\zeta &\sim \mathcal{U}(-0.1, 0.1), \\ C_e &= \mathbf{0}, & D_{e\omega} &= \mathbf{0}. \end{aligned} \quad (21)$$

As final part of the noise model initialization, the components of  $D_{\zeta\omega}$  are initialized using the same procedure as used for the initialization of  $D_{zw}$ . Lastly, to initialize the parameters of the scheduling map, represented by a ResNet, uniform Xavier initialization (Glorot and Bengio (2010)) is used. Bias terms are initialized as zero.

## 6. SIMULATION EXAMPLES

To evaluate the performance of the identified models, we consider two simulation examples. First, a *Nonlinear Mass-Spring-Damper* (NL-MSD) system is considered to evaluate the benefit of rational-scheduling dependency and to display the noise modelling capability of the method. Secondly, we consider a *Control Moment Gyroscope* (CMG), previously used to evaluate performance of joint-estimation methods, enabling comparison of the proposed methodology to existing methods. As a performance measure, the *Best Fit Rate* (BFR) is considered, it is defined as:

$$\text{BFR} = \max \left\{ 1 - \frac{\sum_{k=0}^{N-1} \|y_k - \hat{y}_k\|_2}{\sum_{k=0}^{N-1} \|y_k - \bar{y}\|_2}, 0 \right\} \cdot 100\%, \quad (22)$$

where  $\hat{y}_k$  is the estimated output and  $\bar{y}$  is the sample mean of  $y$ .

The scheduling maps of each model are parametrized as a 2-hidden layer ResNet with 6 neurons per hidden layer and tanh activation functions. The LPV-LFR models of both examples are trained according to the same optimization strategy. Each model is trained for 1000 Adam epochs followed by 4000 epochs using L-BFGS. Leveraging the efficiency of the JAX-based method, all models are trained for 100 random initializations, the model with the highest BFR on validation data is subsequently selected. All models are trained on an AMD Ryzen 4800H CPU.

### 6.1 Nonlinear Mass-Spring-Damper

To illustrate the trade-off between affine and rational-dependency LPV-LFR models we show a numerical example based on a DT NL-MSD system described by (23):

$$\begin{aligned} x_{1,k+1} &= x_{1,k} + T_s x_{2,k}, \\ x_{2,k+1} &= x_{2,k} + \frac{T_s}{m} \left( u_k - \frac{3}{5} x_{1,k} u_k - k_1 x_{1,k} - k_2 x_{1,k}^3 - d_1 x_{2,k} \right), \\ y_k &= x_{1,k}, \end{aligned} \quad (23)$$

with  $T_s = 0.1$ ,  $m = 1$ ,  $k_1 = 0.1$ ,  $k_2 = 1$ ,  $d_1 = 1$  and where  $x_1$  represents the position of the system and  $x_2$  its velocity. Additionally, to illustrate the capability of learning nonlinear noise models we generate filtered noise using the following noise model:

$$\begin{aligned} s_{k+1} &= \left( \frac{277}{300} + \frac{2}{30} x_{1,k} \right) s_k + \left( \frac{277}{300} - 0.65 + \left( \frac{2}{30} - \frac{1}{4} \right) x_{1,k} \right) e_k, \\ v_k &= s_k + e_k. \end{aligned} \quad (24)$$

This plant and noise model feature nonlinearities based on  $x_1$  and  $x_1^2$ , theoretically requiring two scheduling variables for affine LPV representations or one scheduling variable for rational dependency models.

*The Data:* Using the DT system we generate three datasets, one training, validation and test set. Each dataset is generated by exciting the system (23) with a different realization of a random phase multisine signal. The training and validation input signals are  $N = 6 \cdot 10^3$  samples long, the test set is longer, consisting of  $N = 3 \cdot 10^4$  samples. Each input has frequency components from  $\frac{1}{6}$  Hz to 5 Hz. The input signals have amplitudes with standard deviation  $\sigma_u = 4$  N. The resulting outputs are subsequently contaminated with filtered noise generated by exciting the system (24) with white noise signals  $e \sim \mathcal{N}(0, 4.05 \cdot 10^{-3})$  resulting in datasets with SNR of approximately 20 dB.

*Model Settings:* To compare the performance of rational and affine-dependency LPV-LFR models we train rational and affine OE models using the well-posed parameterization with  $(n_x = 2, n_p = 1, \eta = 3)$ , matching the required order necessary for rational models. Subsequently we train affine OE models with  $(n_x = 2, n_p = 2, \eta = 1)$  matching the theoretically required order for affine models. Afterwards, each type of OE model is used to initialize a BJ model ( $n_s = 1$ ) to additionally identify the noise process.

*Results:* The results achieved on the NL-MSD benchmark are shown in Table 1. The results show that all considered models are capable of achieving accurate results on training and validation datasets, however, when evaluated on the longer test set, the affine model ( $n_p = 1$ ) fails to generalize well, as it is not able to capture the true underlying system. This is also illustrated in Figure 1, where it can be observed that the affine model residuals have higher peaks than its rational counterpart, as the rational model generalizes better. Furthermore, the results show that nonlinear noise models can accurately be captured from the data if models of sufficient scheduling order are used.

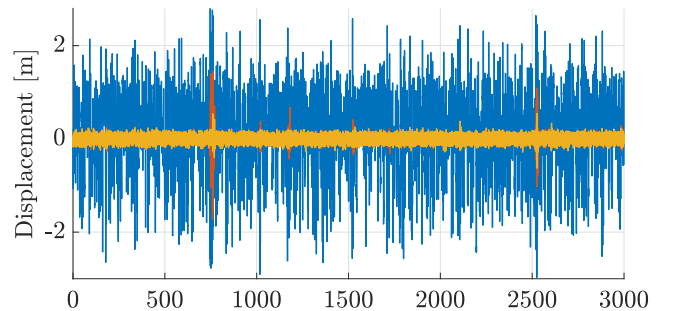


Fig. 1. True output (—), affine LPV-LFR-BJ ( $n_p = 1$ ) error (—) and rational LPV-LFR-BJ error (—) on the test data of the NL-MSD.

Table 1. Results on the NL-MSD example using LPV-LFR models.

| Model              | BFR train | BFR val | BFR test | type | time  |
|--------------------|-----------|---------|----------|------|-------|
| True System        | 89.37     | 89.77   | 89.95    | OE   | -     |
| True System        | 91.65     | 92.16   | 92.05    | BJ   | -     |
| Affine ( $n_p=1$ ) | 89.66     | 89.40   | 80.71    | OE   | 64 s  |
| Affine ( $n_p=2$ ) | 89.57     | 89.75   | 84.57    | OE   | 64 s  |
| Affine ( $n_p=1$ ) | 91.53     | 91.23   | 87.80    | BJ   | 100 s |
| Affine ( $n_p=2$ ) | 91.52     | 91.99   | 91.82    | BJ   | 122 s |
| Rational           | 89.59     | 89.67   | 83.73    | OE   | 161 s |
| Rational           | 91.52     | 91.92   | 91.41    | BJ   | 292 s |

## 6.2 Control Moment Gyroscope

As second simulation example, we consider the 3-DOF CMG and compare achieved performance with the methods presented in Verhoek et al. (2022) and Bemporad and Tóth (2025). The 3-DOF CMG, as shown in Figure 2, consists of a disk mounted inside a blue gimbal, which in turn is mounted inside a red gimbal which is mounted inside of silver gimbal. The disk, and gimbals are each equipped with a DC motor for actuation and an encoder to measure the angular positions ( $q_1, q_2, q_3$  and  $q_4$  for the disk, blue, red, and silver gimbal respectively). During the simulation the red gimbal is locked at  $q_3 = 0$ , the blue gimbal  $q_2$  is actuated with a motor current  $i_2$ , and the angular velocity of the silver gimbal  $\dot{q}_4$  is measured and considered as output. The velocity of the disk  $\dot{q}_1$  is controlled independently, tracking a random multi-level-reference signal with amplitudes between 30 and 50 rad/s and dwell time of 4-8 seconds. The resulting dynamics can, for small angles of  $q_2$ , be interpreted as a nonlinear SISO model between the input  $i_2$  and output  $\dot{q}_4$  with dynamics dependent on the external input  $\dot{q}_1$ .

*The Data:* As Verhoek et al. (2022) describes, the input of the first gimbal is excited with a sinusoidal carrier wave with white noise added:  $u_k = \frac{1}{2} \sin(\omega T_s k) + v_k$  where  $v_k \sim \mathcal{N}(0, \sigma_v^2)$  and  $\sigma_v = \frac{1}{3}$ . For each dataset, the frequency of the sinusoidal carrier wave is drawn from a uniform distribution,  $\omega \sim \mathcal{U}(1, 2)$ . Furthermore, an OE noise structure is considered with error signal  $e_k \sim \mathcal{N}(0, \sigma_e^2)$  with variance  $\sigma_e^2 = 2.2 \cdot 10^{-5}$  corresponding to a SNR of 35 dB. The velocity of the disk follows the previously specified reference. The data is generated using CT simulation with RK4 integration and is subsequently down-sampled to 100 Hz. A training set consisting of  $10^4$  samples is created as well as a validation set of  $3 \cdot 10^4$  samples.



Fig. 2. Picture of CMG considered in simulation.

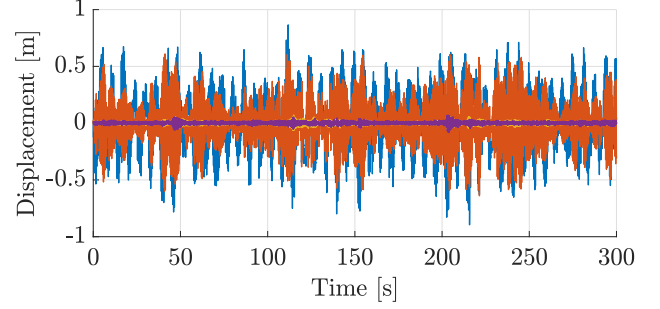


Fig. 3. True output (—), LTI error (—), affine LPV-LFR error (—), and rational LPV-LFR error (—) of the validation data on the CMG.

*Model Settings:* To compare the LPV-LFR model structure with the methods described in Verhoek et al. (2022); Bemporad and Tóth (2025), we opt to train models with the same state and scheduling order ( $n_x = 5, n_p = 3$ ). We set  $n_s = 0$  to obtain an OE noise model, matching the data-generating system. Rational-dependency and affine-dependency-based models are trained for comparison, both with  $\eta = 1$ .

*Results:* The simulation performance of an LTI, affine LPV-LFR and rational LPV-LFR model are shown in 3 and compared with the results in Bemporad and Tóth (2025) in Table 2. The results show that for affine models results similar to existing joint-estimation methods can be achieved, for rational models however, performance is slightly worse.

## 7. CONCLUSION

This work has proposed the use of the linear fractional representation for LPV models in the joint-estimation of LPV models and scheduling maps. A novel parameterization which, by construction, guarantees well-posedness of LPV-LFR models with rational scheduling dependency, has been introduced. The well-posed parameterization has been shown to be capable of modelling nonlinear models and noise processes accurately on a simulation example. Furthermore, the well-posed parameterization has been compared with affine LPV-LFR models and shown to provide more accurate modelling capability with equivalent scheduling order. Finally, affine and rational-dependency LPV-LFR models have been compared to state-of-the-art joint-estimation methods for LPV models on a benchmark where LPV-LFR models have been shown to achieve results close to existing methods.

Table 2. Comparison of LPV identification results on the CMG dataset. Values of true model, LPV-SS and LPV-SUBNET are taken from Bemporad and Tóth (2025) LPV-SUBNET method is described in Verhoek et al. (2022). Comparison performed on different hardware, hence computation times may differ.

| Model          | BFR train | BFR val | type | time           |
|----------------|-----------|---------|------|----------------|
| True System    | 98.16     | 98.19   | OE   | -              |
| Affine LPV-LFR | 97.32     | 96.37   | OE   | 7 min 8 s      |
| Rat. LPV-LFR   | 97.08     | 95.86   | OE   | 11 min 4 s     |
| LPV-SS         | 97.61     | 96.50   | OE   | 47.54 s        |
| LPV-SUBNET     | 97.28     | 96.40   | OE   | $\approx 10$ h |

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